Knight's Tour

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1. Knight's Tour

The knight is an unusual chess piece. It moves in an "L" shape, once in one direction and twice in a perpendicular direction:

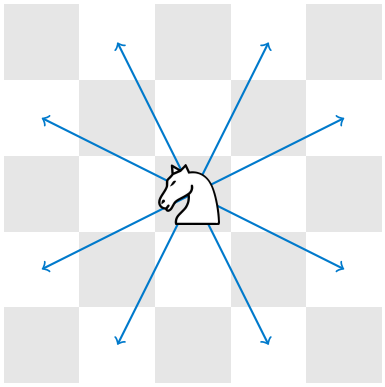


Figure 1: A knight and its possible moves.

The **Knight's Tour** is a sequence of moves where a knight visits each square of a chessboard exactly once. To analyze it, we can model the chessboard as a **graph**, $G = (V, E)$. In this graph, the **vertices** V represent the set of $m \times n$ squares on the board, and an **edge** E connects two vertices if and only if a knight can move between them.

In graph theory, we have a few terms that can help us talk about this tour: A **walk** is a series of vertices where each vertex is connected to the next by an edge. A knight travelling across a chessboard is a walk. A **path** is a walk that never visits the same vertex twice. The Knight's tour is a special kind of path: it visits every vertex in V exactly once. In graph theory, this is known as a **Hamiltonian path.** A "closed" tour, where a knight can return to its starting square, is a **Hamiltonian cycle.** All other tours are "open."

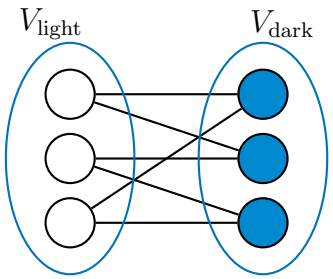


Figure 2: A bipartite graph with disjoint vertex sets V_{light} and V_{dark}.

Because the knight always moves from a light square to a dark square (or vise versa), our graph G is **bipartite**: We can partition the vertices into two disjoint sets V_{light} and V_{dark}, where every edge in E connects a vertex in V_{light} to one in V_{dark}, but no vertex connects to one in the same set.

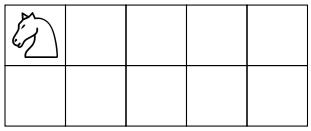
2. Restrictions

2.1. Disconnection

The Classical Knight's Tour considers an $m \times n$ chessboard. We can see very quickly that a Knight's tour cannot exist on a $1 \times n$ or $2 \times n$ chessboard: the Knight cannot possibly move enough to return to its starting square.



1×5 board, Knight just can't move



2×5 board, Knight can only backtrack

Figure 3: A Knight's Tour is impossible on these boards.

On a $1 \times n$ board, the graph G has n vertices but zero edges! Since the knight cannot move from one vertex to another, no path exists, and no tour is possible. (Except for the trivial 1×1 case.) On a $2 \times n$ board, the graph is **disconnected**! A knight on $(1, 1)$ can only reach squares in one **component** of the graph, while a knight starting on $(2, 2)$ can only travel the other component. Since the graph is disconnected, no path exists between all vertices, no Knight's Tour is possible.

2.2. Bipartite Coloring

Recall that the chessboard is a bipartite graph: The knight also alternates black, white, black...

Theorem 1

A closed knight's tour can only exist on a board with an even number of squares.

Proof. For a Knight's Tour to be closed, it must return to its starting square, meaning there must be an equal number of light and dark squares. When the dimensions of the chessboard are both odd, this is not possible, and therefore a closed knight's tour is impossible on any $m \times n$ board where m and n are odd. □

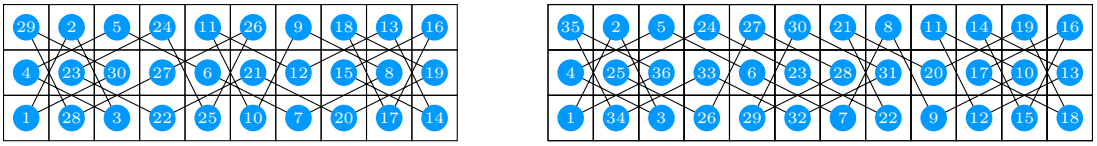


Figure 4: Euler thought closed tours were impossible for boards with sides < 5, but Ernest Bergholt found 3×10 and 3×12 tours in 1917.

2.3. Open Tours

Well, what about paths? An open tour on an odd board still needs to alternate colors, meaning an open tour is possible only if it starts and ends on the majority color.

If we consider a 3×3 board, we can see that there is no way for the knight to get to the middle square, and there is no way for the knight, once starting in the middle, to get to any other square. So, the square in the middle is disconnected from the graph, and no Hamiltonian path exists. This rules out $3 \times \{1, 2, 3\}$, but we can see that a Hamiltonian path is possible on a 3×4 grid!

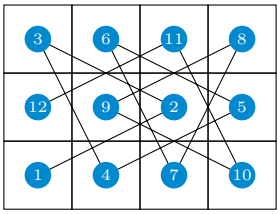


Figure 5: A Knight's Tour on a 3×4 chessboard.

A 3×4 grid has 12 squares: an even number. Our reasoning fails to rule out a closed tour, but a closed tour is still impossible. While bipartite coloring is a necessary condition for a closed tour, it is not sufficient to show that one exists.

2.4. 4 × n

Theorem 2

No closed Knight's Tour exists on a $4 \times n$ board.

Proof. We can justify this with a coloring argument. We can color the squares as follows, with A, B alternating on the top and bottom, and C, D alternating in the middle:

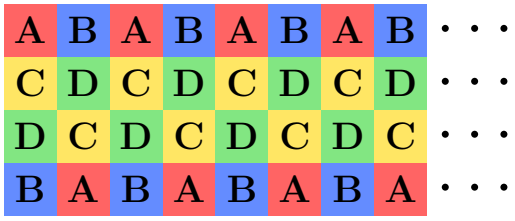


Figure 6: Each row has one of A, B, C, D.

There are an equal number of squares labeled A, B, C, D as this is true for each row. For a closed Knight's Tour to exist, we don't need to consider any specific starting point, since the knight goes to and comes from every square, so if it fails to go through one and come back to it, no Knight's Tour exists.

Thus, we assume the knight starts at square A. From here, it must go to a square C. However, it must have also *come* from a square C. This means that when it jumps from a C to a D cell, it *must* jump back to a C cell to get to another A cell. An example sequence may look like:

$$A \to C \to D \to \boxed{\text{C}} \to A \to C...$$

There are an equal number of A and C cells in the grid, and to form a closed Knight's Tour we must jump to all of them. However, this implies that we must travel to all of the A and C cells before any of the B and D cells, because otherwise we would count more C cells than A cells.

Then, we must finish the tour by traveling from D to B, back and forth. Since a B only leads to a D, this final D must connect back to an A. But a D cannot connect back to an A without travelling through a C. Thus, the tour can never be closed, so, no knight's tour exists on a $4 \times n$ board! □

2.5. Schwenk's Theorem

Schwenk's Theorem (1991)

For an $m \times n$ chessboard (with $m \leq n$), a closed Knight's Tour exists if and only if all of the following conditons are false:

- m and n are both odd.
- $m = \{1, 2, 4\}$.
- $m = 3$ and $n = \{4, 6, 8\}$.

Proof. We have already proven that a Knight's tour cannot exist if (a) or (b) are met. We

have also proven that a closed Knight's tour cannot exist on a 3×4 board.

An important feature of a graph with a Hamiltonian cycle is that removing k vertices from G can leave at most k connected components

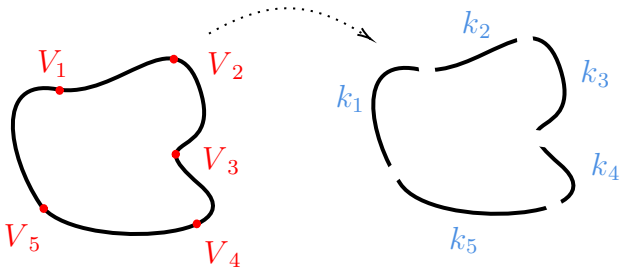


Figure 7: Removing $V_1 ... V_5$ leaves $k_1 ... k_5$ as parts.

Consider the board 3×6. Let's remove the vertices $(3, 1)$ and $(3, 3)$. If G contains a Hamiltonian cycle, the board should split into at most 2 components. However, we can see that the square $(1, 2)$ is disconnected, creating one component. Likewise $(5, 2)$ is disconnected. The other 14 squares remain connected to each other. Clearly, the board has split into 3 components, and thus a Hamiltonian cycle cannot exist!

We know now that a Knight's tour is impossible on the boards that satisfy the conditions of Schwenk's theorem. Additionally, we can construct every other board to prove they have Knight's tours! The full proof for this is too long to discuss here, but the construction is fairly similar for each case:

$(1,3)$	$(2,3)$	$(3,3)$	$(4,3)$	$(5,3)$	$(6,3)$
$(1,2)$	$(2,2)$	$(3,2)$	$(4,2)$	$(5,2)$	$(6,2)$
	$(2,1)$	$(3,1)$	$(4,1)$	$(5,1)$	$(6,1)$

$(1,3)$	$(2,3)$		$(4,3)$	$(5,3)$	$(6,3)$
$(1,2)$	$(2,2)$	$(3,2)$	$(4,2)$	$(5,2)$	$(6,2)$
$(1,1)$	$(2,1)$		$(4,1)$	$(5,1)$	$(6,1)$

Figure 8: A chessboard with two squares removed.

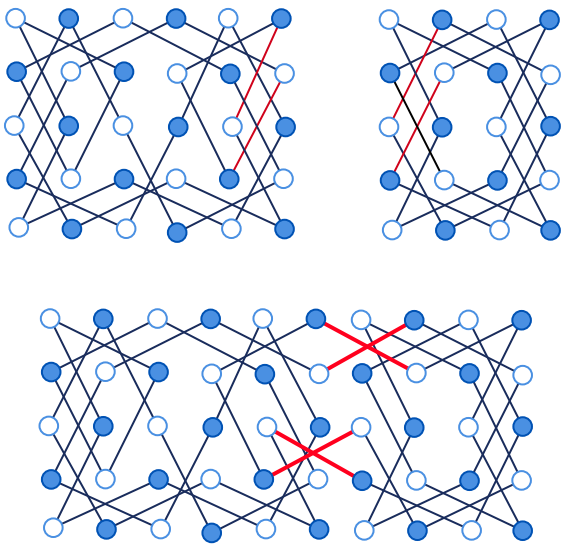


Figure 9: Two edges are cut, and then stitched together to create a larger Knight's Tour.

First, take a known tour, say, the 5×6 tour. Then, attach on a 5×4 block of squares. We can "cut" two specific edges on the 5×6 tour, right at the border, and do the same for the 5×4 block. We can then stitch them together to create a 5×10 tour. Note that the same process can be applied to the 5×10 board, and so on!

Note that the patterning on the completed board resembles the ending pattern of the original boards, meaning we can continuously stitch these boards together. The same argument can be applied to columns. Closed knight's tours can be constructed for the following pairs (m, n) although we will not explicitly show it here:

$$(3, 10), (3, 12), (5, 6), (5, 8), (6, 6), (6, 8), (7, 6), (7, 8), (8, 8)$$

From each of these pairs, by the stitching argument we described earlier, we can add 4 to any coordinate in the pairs above any number of times we want. By induction with these base cases, we get every pair that satisfies the conditions of Schwenk's Theorem, so closed Knight's tours are possible are on all of these chessboards. □

3. Variants

3.1. Holey Boards!

What if the chessboard isn't empty? In our proof that a knight's tour is impossible on a 3×6 board, we considered removing squares of the chessboard. This is the same as removing a vertex from a graph. If we place a knight on $(3, 3)$, we are no longer looking for a Hamiltonian cycle on G but on G', where the vertex $(3, 3)$ and all of its connected edges have been deleted.

While this can make tours impossible, it can make new tours possible! The smallest Knight's Tour on a 3×3 board only exists with 2 cells removed, while the smallest closed tour exists on a 3×3 board with just the center square removed.

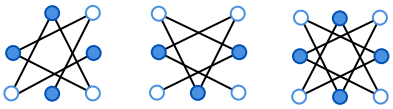


Figure 10: Two open Knight's Tours and one closed Knight's Tour.

Finding the second formation, the U-shape, appeared in one of the earliest Arabian chess manuscripts as a puzzle: the trick is to start a Knight's jump away from the empty non-central square!

3.2. ∞ Board

On an infinite board, there are no corners or edges to restrict the knight. Obviously, thinking about a closed knight's tour on an infinite $\mathbb{Z} \times \mathbb{Z}$ board is a little silly, but we can ask two other questions:

(a) Can our knight, starting at $(0, 0)$, reach any square (m, n)?

(b) Can we find a path that reaches every square exactly once?

For (a), we only need to prove that the Knight can move to $(0, 1)$ and $(1, 0)$. In other words, if we can find a sequence of moves that leaves us with a net $+1$ to the x and y coordinates of the knight, we can apply this sequence repeatedly to reach (m, n). We can easily find this sequence, after a few moves:

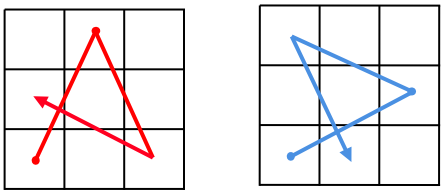


Figure 11: The path a Knight walks to reach $(0, 1)$ and $(1, 0)$.

...and so we can always reach (m, n). (We can do a similar procedure to achieve every negative ordered pair as well).

For (b), we can construct a tour that extends infinitely:

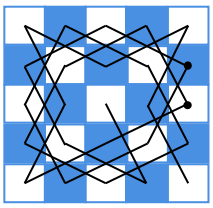


Figure 12: Infinite Knight's Tour!

This knight's tour, when repeated, will reach every square exactly once.

3.3. (a, b) Knight

Instead of changing the board, what if we change the *knight*? We can call our current knight a $(1, 2)$ knight, meaning it moves 1 step in one direction, and twice in a perpendicular direction. We can generalize this to an (a, b) knight, moving a in one direction, and b in a perpendicular direction. One question we can ask is: which (a, b) knights can reach every square on an infinite board?

The set of all possible positions a knight can reach forms a subgroup P of the lattice $\mathbb{Z} \times \mathbb{Z}$. This subgroup can be generated by 8 move vectors, representing the 8 moves of the knight:

$$P = \langle (\pm a, \pm b), (\pm b, \pm a) \rangle.$$

Can the knight reach every square (m, n)? is equivalent to asking: does $P = \mathbb{Z} \times \mathbb{Z}$? Similar to before, we can prove this by showing that the knight can make a net move of $(1, 0)$ and $(0, 1)$. It can then reach any point (m, n) by combining this sequence of moves.

First, $\gcd(a, b)$ must be 1 in order for the knight to reach every square. In other words, a and b must be coprime. Suppose otherwise. If $\gcd(a, b) = d > 1$, every coordinate a knight can reach from $(0, 0)$ is a multiple of d, which means not every coordinate can be reached.

There is another restriction on the value of a and b. If we color an infinite board in a checkerboard pattern, we need the knight to switch the color it's on every move. However, if $a+b$ is even, then the knight will continue to be on the same color. Therefore, it is necessary that a and b have different parity, or in other words, $a \not\equiv b \pmod 2$. In fact, this, along with a and b being coprime is sufficient for the (a, b) Knight to be able to traverse an infinite board.

To show that this condition is sufficient, suppose that $\gcd(a, b) = 1$ and a is odd while b is even. By Bezout's Lemma or the Extended Euclidean Algorithm, we can find integers c and d such that $ac + bd = 1$. In fact, we claim we can find a solution where c is odd and d is even. c is odd no matter what since if we reduce this equation mod 2, we see $ac \equiv 1 \pmod 2$, and a being odd implies c is odd. On the other hand, if d is odd, we can construct the new solution $a(c - b) + b(d + a)$. Here, $c - b$ is still odd, but since a and d are both odd, $d + a$ is even. Thus, we can just assume c is odd and d is even. The idea now is to construct a sequence of moves that will take the Knight to the square $(1, 0)$ based on the coefficients c and d. c will represent the net number of a-moves in the x-direction (# moves a to the right - # moves a to the left) while d will represent the net number of b-moves in the x-direction. If we want the net number of a-moves in the x-direction to be c, we can take the total number of a-moves to be any odd integer greater than or equal $|c|$; suppose we set the total to some large odd integer N. We have to choose the y-direction of each these moves to either be $+b$ or $-b$. We will choose a of them to be $-b$ and the rest (which come in pairs) to have an equal number of $+b$'s and $-b$'s. This makes the net movement in the y-direction due to these moves $-ab$.

Similarly, we can set the net number of b-moves in the x-direction to be any even integer greater than or equal to $|d|$, say M. We have to choose the y-movement for each of these moves to be either $+a$ or $-a$. We will choose b of them to be $+a$ and the rest to come in pairs of $+a$ and $-a$. Thus, overall, these moves contribute a total movement in the y-direction of

$+ab$.

Putting both of these types of moves together–ones where we move a in the x-direction and ones where we move b in the x-direction–we have a net movement in the x-direction of $ac + bd = 1$ and a total net movement in the y-direction of $-ab + ab = 0$. Thus, the Knight can take a series of moves to get to $(1, 0)$.

To summarize, since we can always construct a sequence of moves that results in a net move to $(1, 0)$ (and by symmetry, $(0, 1)$), the (a, b) knight can reach every square in $\mathbb{Z} \times \mathbb{Z}$ iff $\gcd(a, b) = 1$ and $a \not\equiv b \pmod 2$.

□

4. Boards all Glued Together

4.1. Cylinders and Tori

Imagine now that we take a chessboard and glue its sides together.

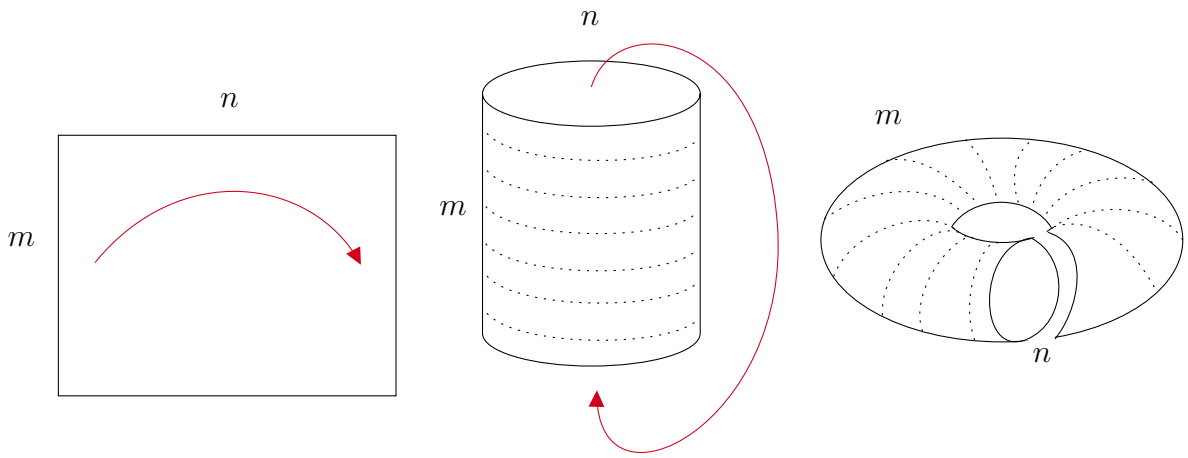


Figure 13: Construction of mxn cylinder and mxn torus from mxn rectangle chessboard.

We can see that an $m \times n$ rectangle and torus are identical to those that are $n \times m$. However, there is a distinction between an $m \times n$ cylinder and a $n \times m$ cylinder. If we take the left edge and "glue" it to the right edge, as in the diagram, we create an m-high cylinder with circumference n. Note that the "row" moves of the knight are the same as when the cylinder was unwrapped, but the "column" moves are repeated mod n.

Consider a knight on a rectangular board, on the far right edge, at square $(a, n-1)$. (Here, we are indexing at 0). The knight wants to move to $(a + 1, n + 1)$, but this is impossible due to the constraints of the board. However, if we wrap the rectangle to be a cylinder, the knight can wrap around the edge of the cylinder and end up at $(a + 1, 1)$. In general, when moving, we take the Knight's y-coordinate mod n. For an $n \times m$ cylinder, imagine rolling and sticking the top and bottom boundaries of the chessboard together. In this case, the column moves are unchanged, but the row moves wrap around.

For the $m \times n$ torus, we can imagine taking our cylinder and having *its* edges wrap

around. In this way, the torus chessboard has no boundary, and the knight can wrap around forever. We can take the new row mod m and the new column mod n to pinpoint a location. The graph G representing the torus (for every $m, n > 4$) is an 8-regular graph. A **regular** graph is one where each vertex has the same degree. This removes the edge restriction, and makes Knight's Tours easier to find.

4.2. Constructing Algorithms

Cylinder Construction

Consider an $m \times n$ cylinder (height m, circumference n). If there exists some Knight's Path of length $m+1$ that starts and ends at the bottom row, visiting every other row exactly once and has a net displacement of $(1, 0)$, then a closed Knight's Tour exists on the cylinder.

Proof. In essence, if we can start on the bottom row and can construct a Knight's Path that visits every row exactly once before returning to the bottom row, we can repeat this algorithm n times to cover every square of every row! Since the knight ends with a net movement $(1, 0)$, the row will be shifted exactly once over, so the knight will travel to every square. Also, since the knight ends by moving to square $(n+1, 0)$, it returns to its starting square, completing a closed Knight's Tour. □

4.3. Proving Existence

Let's try our coloring argument again. For a closed tour on a bipartite graph, the number of dark squares must equal the number of light squares. If mn is odd, $V_{light} \neq V_{dark}$. So, we may be tempted to say that no knight's tour exists on a torus where mn is odd, but this is false! Consider a square of a chessboard $(m-1, c)$, with m and c both odd. Let a square (a, b) with $a+b$ odd be dark and a square with $a+b$ even be light. Since the net movement of this knight is odd, the knight moves from odd to even net coordinates on its bipartite graph (and vice versa), thus moving from dark to light squares. A possible move for the knight is $(m+1, c+1)$. This seems to obey our principle: $m+1$ is even and $c+1$ is even, so the knight has moved from a net-odd square to a net-even square. However, as row $m-1$ is on the edge of the rectangle, $m+1$ is row 1. Thus the knight is on $(1, c+1)$, which is a net-odd/dark square. Because the knight can move from a dark square to another dark square, this checkerboard pattern isn't valid doesn't make the chessboard bipartite.

A similar argument can be used for the $m \times n$ cylinder. Consider square (m, c). Here, rows repeat mod m. The knight can move to $(1, c+1)$, creating a net move from an odd square to an odd square.

□

So, what restrictions *do* exist on bent boards?

Theorem 3

For any cylinder not of size $4 \times n$, $2 \times n$, or $1 \times n$, there exists a closed knight's tour.

Proof. We can prove the existence of a knight's tour on a $3 \times n$ board by construction, as demonstrated by Kelley Seibel in *The Knight's Tour on the Cylinder and Torus.*

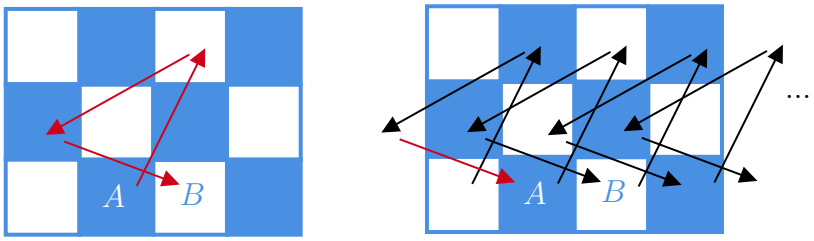


Figure 14: An algorithm for a closed Knight's Tour on a $3 \times n$ cylinder.

We start from A and we manuver to B in the above manner. In this way, we reach a square of each row exactly once per iteration, and we end at the square one to the right of A. If we continue this pattern around the cylinder, we loop around and end a knight's jump away from A, completing the Knight's tour. This closed Knight's Tour is constructed indistinguishably no matter the parity of the board, so there is no parity restriction. □

For boards $5 \times n$ and larger, (n here is ≥ 5), we know that a closed tour always exists on a cylinder where at least one of m, n are even, because they exist on an $m \times n$ board. So, we need only consider the case where both are odd. We can construct an algorithm by starting at a square and repeating the following sequence:

$$(+1, +2), (-1, +2)$$

until we reach the top of the board, at move $\frac{m-1}{2}$, because m is odd. From here, we can move $(-2, -1)$ and repeat the same sequence moving down, hitting the rows we missed previously. This sequence will always end up a knight's jump away from a square one unit to the right of our starting square, as the net move will be $(+1, 0)$.

To see this, we can consider when m is 1 mod 4 and when m is 3 mod 4. When m is 1 mod 4, the net move when going up the board is $(m, 0)$, because the arrows reach the top square, and return to the "color" of the starting square. (We can see this by observing $m = 5$, and seeing that the sequence repeats mod 4.) When going down, we will move net $(-m, 1)$. We can see that, in the x, the final arrow does not cancel, creating a net movement right. So, our total move vector is $(0, 1)$.

We can see a similar thing for when $m \equiv 3 \mod 4$. In this case, the net x move occurs when traveling up, creating a net $(m, 1)$ move. Then, when it goes down, it creates a $(-m, 0)$ move, totaling a net $(1, 0)$ move.

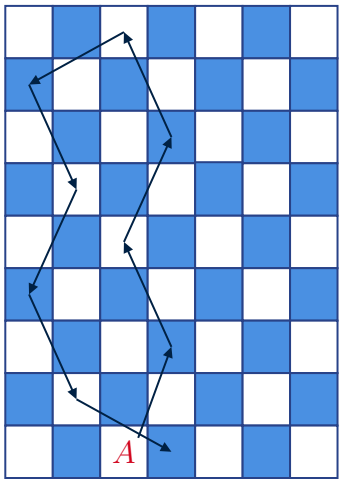


Figure 15: An example of the above algorithm on a 9×7 cylinder.

□

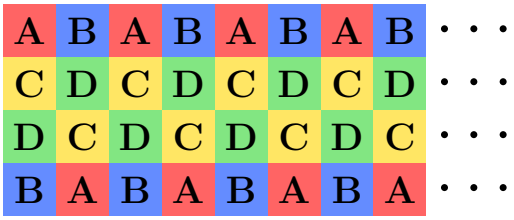
Theorem 4

No closed knight's tour exists on any $1 \times n$ cylinder, or $2 \times n$, $4 \times n$ cylinders with n even.

Proof. As with the $1 \times n$ board, the knight has no squares to travel to, so (besides the trivial 1×1 cylinder), there can exist no closed Knight's Tour on a $1 \times n$ cylinder.

For the $2 \times n$ board, with n even, a knight can move $(\pm 2, \pm 1)$ but not $(\pm 1, \pm 2)$. This means that the parity of the knight's x coordinate is unchanged after a knight's move. So, if the knight starts on an even numbered column, it can only reach other even numbered columns. When a knight reaches square $x = n + 1$ it goes to square $x = 1$, preserving parity under wrapping. Thus, the knight can only reach squares with columns numbered of its starting parity, and can never complete a full knight's tour.

Lastly, for the $m \times n$ cylinder, we can use an identical argument as we used for the board. To bring back our diagram from before:



We can split the chessboard into sections A, B, C, D, with A, B having degree 4 and opposite column parity, and C, D having degree 6 and opposite column parity. Each group A, B, C, D is equal in size, as there is one of each per row. Assume we start at square A. From here, even with wrapping, we must go to a square C, and we must have come from a square C. Thus, there are two $C's$ for one A. In order for the amount of C and A to be equal, we must then go to every C and every A before any B or D. But then, once we get

to B, D, there is no way to get back to A or C to complete the tour, as it would disrupt the balance of A and C. Thus, no closed knight's tour can exist on a $4 \times n$ cylinder, where n is even. □

For the $2 \times n$ cylinder, with n odd, we can construct a tour by considering the cylinder in 4 "parts". (Here, I quote "parts" because they really are all connected). We can start on the top left corner of the unwrapped cylinder, and number each column, indexed at 0. We can have our knight continuously move $(+2, \pm 1)$, starting with -1 and alternating each turn. Consider the net-xy move k, l. When k exceeds n the first time, k is set to 1 and our column parity is switched from even to odd. We can then continue to apply $(+2, \pm 1)$ until we reach $k = n$, and when this happens, k is set to 0.

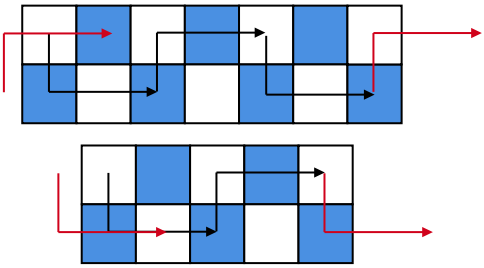


Figure 16: Knight's Tour on 2×7 and 2×5 cylinders.

When $n + 1 \equiv 2 \mod 4$, the row parity will switch during the first iteration, and will reach -1, not 0, as its net movement, before wrapping. There are say, j moves before the first wrapping, and there will be j moves before the second wrapping, so there will be $j + 1$ to complete the second wrapping. Because of this, the knight will have a net -1 row move by the end of the second iteration, and the knight will be on the second row, on an odd numbered column. It will continue to move forward, in the pattern $(+2, \pm 1)$, filling each square it hadn't before.

When $n + 1 \equiv 0 \mod 4$, this does NOT switch the row parity. However, before the end of the second wrapping, there will be exactly one column of space, meaning that for the next two iterations, the Knight will be on the 2nd row.

We move a total of $2n$ moves, and because there are an even number of $+1$s and -1s, we have a net 0 row move. Also, we shift our x-coordinate by $4n$, which is 0 mod n. Thus, after we finish this procedure, we return to our starting square and complete the closed Knight's tour.

For the $4 \times n$ algorithm, we can make use of the fact that the column parity changes when wrapping. We can consider the following algorithm:

... | 3 | | | | ...
... | | | 2 | | ...
... | | 4 | | | ...
... | | 1 | | 5 | ...

The algorithm is successful, as it reaches each row exactly once before reaching the

bottom row. However, it is 2 squares away instead of exactly one. However, if column parity switches when wrapping, we can first hit every odd column, then the even ones, and, after the second wrapping, we finish the circuit on the odd square we started on.

Theorem 5

A closed Knight's tour exists on *every* torus.

Proof. Note that we only need to prove the cases $1 \times n$ for all n and $2 \times n, 4 \times n$, with n even, as the cylinder covers all other cases. (The torus is just a cylinder with more options, as a cylinder is a board with more options).

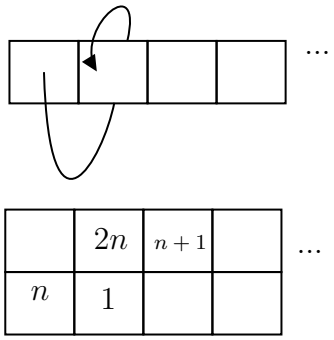
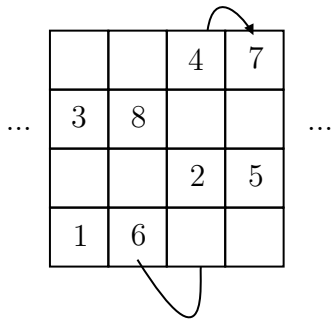


Figure 17: A Knight's Tour on $1 \times n$ and $2 \times n$ tori.

For the $1 \times n$ case, we can move once horizontally and twice wrap vertically, ending on the square directly to the right of the starting square. From here, we can apply this for the whole length of the torus, and we end right back at the starting square.

For the $2 \times n$ case, we can use the result from this. We can work our way around the first row, then jump to the second row and do the same. And since the ending square of the first row is both once to the left of the starting square and a knight's jump away from the starting squrae of the second row, the knight can always jump back to the starting square, completing the knight's tour.

For the $4 \times n$ case, we can construct the following algorithm:



Since the number of columns is even, no iteration will be incomplete. When $n \equiv 2 \mod$

4, the knight simply wraps during the iteration! The ending jump is a knights jump from the starting move, so we have a closed knight's tour. □

5. Conclusion

The Knight's Tour is an interesting puzzle that has fascinated mathematicians for thousands of years. Because of the Knight's unique style of movement, it's nice to model it on a graph and observe its properties. The Knight's Tour is also pretty huge in computer science, with algorithms existing to find these tours on various boards. For further reading, check out George Jelliss's *Knight's Tour Notes*; it has a lot of cool history and variations of the Knight's Tour!

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